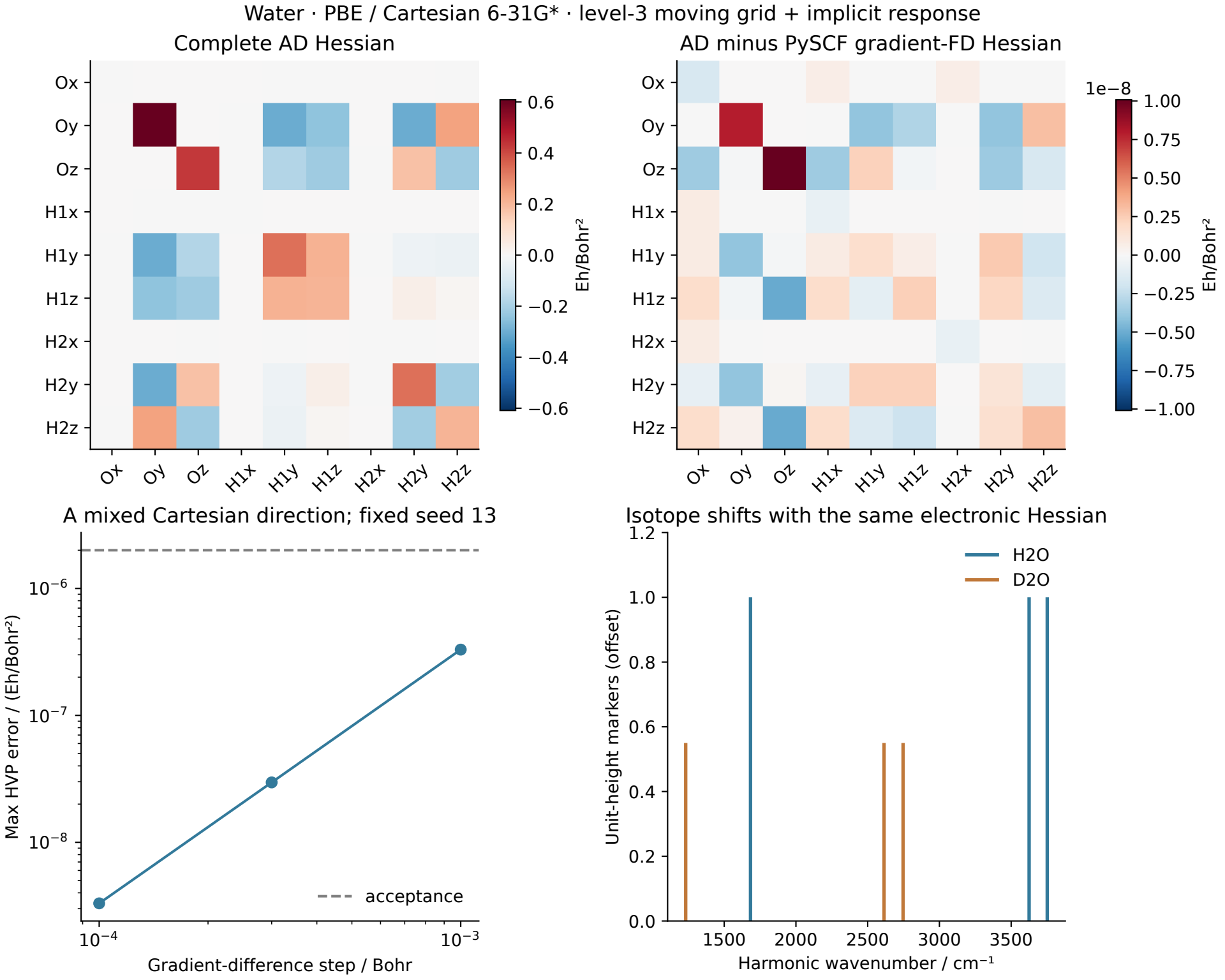
Water · PBE / Cartesian 6-31G* · level-3 moving grid + implicit response
Complete AD Hessian
AD minus PySCF gradient-FD Hessian
Ox
Oy
Oz
H1x
H1y
H1z
H2x
H2y
H2z
Eh/Bohr²
1e−8
A mixed Cartesian direction; fixed seed 13
Max HVP error / (Eh/Bohr²)
Gradient-difference step / Bohr
acceptance
Isotope shifts with the same electronic Hessian
H2O
D2O
Unit-height markers (offset)
Harmonic wavenumber / cm⁻¹